Metacognition
-Term coined by **John Flavell (1970)**
Consists of both metacognitive knowledge and metacognitive experiences.
"Thinking about thinking" or "learning how to learn"
Refers to **higher order thinking** which involves active awareness and control over the cognitive processes engaged in learning
Strategies to Develop Metacognition

1. **Knowledge of Task**
2. **Knowledge of Strategies**
3. **Knowledge of Self**
4. **Planning**
5. **Monitoring**
6. **Evaluation**

Learning- Centered Psychological Principles

1. Nature of Learning Process
2. Goals of the Learning Process
3. Construction of Knowledge
4. Strategic Thinking
5. Thinking about thinking
6. Context of Learning
7. Motivational and emotional influences on learning
8. Intrinsic motivation to learn
9. Effects of motivation on effort
10. Developmental influences on learning
11. Social influences on learning
12. Individual differences in learning
13. Learning and diversity
14. Standards and assessment

Alexander and Murphy gave a summary of the 14 principles and distilled them into five areas

1. The knowledge base
2. Strategic processing and control
3. Motivation and affect
4. Development and Individual Differences
5. Situation or context

Individual differences
-<u>Diversity</u> is everything that makes people different from each other. Factors that makes us different

- **Race**
- **Ethnicity**
- **Gender**
- **sexual orientation**
- **socio-economic status,**
- **Ability**
- **Age**
- **religious belief**
- **or political conviction.**

7 ways to encourage a culture of diversity in your school

1. Re-evaluate your teaching materials
2. Get to know your students
3. Be willing to address inequality
4. Connect with families and community
5. Meet diverse learning needs
6. Support professional development opportunities

Learning and Thinking Styles and Multiple Intelligences

Sensory System

-Sensory Receptors

Specialized cells or multicellular structures that collects information from the environment

-Sensation

A feeling that occurs when brain becomes aware of sensory impulse

-Perception

A person's view of the situation; the way the brain interprets the information

General Senses

-Receptors that are widely distributed throughout the body

-Skin, various organs and joints

Types of Receptors

1. **Thermoreceptors -** Respond to changes in Temperature (heat)
2. **Pressure-mechanoreceptors -** Respond to mechanical forces (movement)
3. **Pain Receptors (Nociceptors)** - Respond to tissue damaged (muscles, joints visceral organs and digestive tract)
   -sprains, bone fractures, burns, bumps, bruises, inflammation (from an infection or arthritic disorder), obstructions, and myofascial pain (which may indicate abnormal muscle stresses)
4. **Special Senses -**Specialized receptors confined to structures in the head (Eyes, ears, nose and mouth)
5. **Chemoreceptors -**Respond to changes in chemical concentration (taste and smell)
6. **Photoreceptors -** Respond to light(eyes)
7. **Mechanoreceptors -** Respond to mechanical forces (balance) Equilibrium; Respond to sound (hearing**)**
8. **Osmoreceptors -**Respond to changes in solute concentration (hypothalamus/carotid artery)

The Eye and Vision

- 70 percent of all sensory receptors are in the eyes

-Each eye has over a million nerve fibers

Parts of the EYE

1. Sclera.
2. Cornea.
3. Iris.
4. Pupil.
5. Lens.
6. Retina.
7. Optic nerves.
8. Aqueous Humor
9. Vitreous Humor
10. Ciliary Body
11. Macula
12. Conjuctiva

EARS Houses two senses

-Hearing (interpreted in the auditory cortex of the temporal lobe)

-**Equilibrium (balance)** (interpreted in the cerebellum)

-Receptors are **mechanoreceptors**

-Different organs house receptors for each sense

Parts of the EAR

External Ear

1. Pinna or auricle
2. External auditory Canal or tube
3. Tympanic Membrane (eardrum)

Middle Ear (Tympanic Cavity)

1. Ossicle
    - Malleus
    - Incus
    - Stapes
2. Eustachian Tube

Inner Ear

1. Cochlea (nerves for hearing)
2. Vestibule (receptors for balance)
3. Semicircular Canal (receptors for balance)

Common EAR Disorders and Diseases

1. Balance Disorders.
2. Cholesteatoma- abnormal collection of skin cells deep inside your ear
3. Ear Infections.
4. Ear Ringing (tinnitus)
5. Hearing Loss.
6. Otitis.
7. Perforated eardrum.

Chemical Senses – Taste and Smell

Both senses use chemoreceptors

- Stimulated by chemicals in solution
- Taste has four types of receptors
- Smell can differentiate a large range of
- chemicals

-Both senses complement each other and respond to many of the same stimuli

Olfaction – The Sense of Smell

-Olfactory receptors are in the roof of the nasal cavity

-Neurons with long cilia

-Chemicals must be dissolved in mucus for detection

Parts of the Nose

1. External meatus.
2. External nostrils.
3. Septum.
4. Nasal passages.
5. Sinuses.

Common Nose Problems

1. Sinus infections (sinusitis)
2. Congestion
3. Airway blockage

The Sense of Taste

1. Taste buds house the receptor organs
2. Location of taste buds
3. Most are on the tongue
4. Soft palate
5. Cheeks

The tongue is made up of three elements

- **Epithelium**
  - **-**Papillae
  - Taste buds
- **Muscles**
- **Glands**
  - **-**Mucous Glands
  - -Serous Glands
  - -Lymph Nodes

Common Tongue Problems

- **Aglossia** absence of the tongue (congenital)
- **Hypoglossia** a short and incompletely formed tongue
- **Burning Mouth Syndrome** also known as glossodynia, glossopyrosis, and stomatopyrosis. It's a chronic condition that can causeTrusted Source a burning sensation on the tongue or elsewhere in the mouth.
- **Macroglossia** refers to when your tongue is larger than what it should be. The condition is also called "big tongue" or "enlarged tongue."
- **Atrophic glossitis** is a condition in which the tongue is missing some or all of its papillae, making its usually rough surface smooth.

Hemisphere of the Brain

Brain one of the largest and most complex organ in the body

-Pruning is the degradation of neurons because of aging

-Plasticity is the ability of the brain to continuously change in response to learning or injury

-The human brain contains 50 billion neurons at birth

-At age 10, children have developed half of the brain cell connections

-Myelination begins prenatally and continues after birth

Brain Anatomy

The human brain is comprised of 3 parts:

- Hindbrain
- Midbrain
- Forebrain

The Hindbrain is located at base of skull and consists of:

- **Cerebellum:** posture, balance, voluntary movement
- **Medulla:** breathing, heart rate, reflexes
- **Pons:** bridge between the spinal cord and the brain
- **Brainstem** is the most primitive part of the brain and controls the basic functions of life: breathing, heart rate, swallowing, reflexes to sight or sound, sweating, blood pressure, sleep, hormonal maturation, and balance.

The Midbrain integrates sensory information

-Handles all sensory information that passes between the spinal cord and the forebrain.

-It is also involved in body movement in relation to auditory and visual signals.

-Located just above the hind brain.

- When viewed in cross-section, the midbrain can be divided into three portions:

  - Tectum (posterior)
  - Tegmentum
  - Cerebral peduncles (anterior)

-contain part of the **substantia negrae**, which (like the ventral tegmental area) contain large collections of dopamine-producing neurons.

The Forebrain contains-

- **Cerebrum:** the largest and most developed part of brain.

-Responsible for intelligence, personality, thinking The cerebral cortex is a gray tissue that covers the cerebrum.

- **The limbic system** is the area of the brain that regulates emotion and memory. It directly connects the lower and higher brain functions.
- **Thalamus:** relay station for all sensory information except smell.
- **Hypothalamus**: controls hunger, thirst, sexual behavior
- **Hippocampus:** important in forming memories
- **Amygdala**: involved in memory and emotions (fear, anger, pleasure, aggression)

LOBES of the Brain

- **Occipital lobe:** for vision and recognition
- **Parietal lobe:** handles information from the senses
- **Temporal lobe:** hearing, memory, emotion, speaking, smelling, tasting, perception, aggressiveness, and sexual behavior.
- **Frontal lobe:** organization, planning, creative thinking

MULTIPLE INTELLIGENCE THEORY BY HOWARD GARDNER

These are:

1. Naturalistic
2. Musical
3. Logical–mathematical
4. Existential
5. Interpersonal
6. Linguistic
7. Bodily–kinesthetic
8. Intra–personal
9. Spatial intelligence.

Learners with Exceptionalities

The arrangement of teaching procedures, adapted equipment's and materials, accessible settings and other interventions designed to address the needs of students with:

1. Learning Differences
2. Mental Health Issues
3. Physical Disabilities
4. Developmental Disabilities
5. Giftedness

INTELLECTUAL DISABILITY: CAUSE

BRAIN MALFORMATION
INFECTIONS - VIRAL
NUTRITIONAL - METABOLIC
DOWN SYNDROME – CHROMOSOMAL

LEARNING DISABILITY

DYSLEXIA

- Developmental disorder characterized by delay and difficulty in:
    - WRITING
    - SPELLING
    - READING

DYSCALCULIA

Poor capacity in mathematics that is not par with age:

1. Arithmetic facts
2. Numerical magnitude
3. Calculations
4. Numbers
5. Patterns
6. Shapes
7. Graphs
8. Charts
9. Directions
10. Measurement
11. Card-board-video Games
12. Schedules

DYSGRAPHIA

1. Impaired Handwriting
2. Difficulty putting ideas into written form
3. Writes slowly or letters go in all directions
4. Trouble holding pencil, pen, crayons, scissors
5. Problems with Punctuations
6. Erases a lot
7. Mix upper-lower case letters
8. Spelling issues

ATTENTION DEFICIT HYPERACTIVITY DISORDER

- ✓ Neurological condition that involves problems with:
  I NATTENTION
  H YPERACTIVITY
  I MPULSIVITY
- ✓ Developmental failure in the Brain circuitry that monitors inhibition and self-control.
- ❖ Cerebral Cortex delayed for about 3 years.
- ❖ Brain Size 5% Smaller
- ❖ Decreased Dopamine
- ✓ Males 3x affected

POSSIBLE CAUSES:

1. GENETIC
2. BRAIN INJURY
3. ENVIRONMENTAL

?. TIME ON SCREENS
?. FOOD

Common Speech Disorders

- ➢ Stuttering
- ➢ Cluttering
- ➢ Apraxia
- ➢ Lisping
- ➢ Articulation Disorders
- ➢ Dysarthria

AUTISM

- ✓ A developmental disability significantly affecting verbal and non-verbal communication and social interaction.
- ✓ Onset is before 3 years old
- ✓ Lifelong Disability
- ✓ Males 4x more affected
- ✓ Cause: Unknown
- ✓ Risk Factors
  - * Genetic
  - * Environmental:
    Lead, Mercury
  - * Parental Age: Over 40

Features of Mental Retardation

-DSM-IV Criteria significantly subaverage IQ (<70)concurrent deficits or impairments in adaptive functioning characteristics evident prior to age 18.

The DSM-IV classifies mental retardation into four stages based on severity:

1. mild (IQ score of 50-55 to approximately 70)
2. moderate (IQ score of 30-35 to 50-55)
3. severe (IQ score of 20-25 to 35-40)
4. profound (IQ score of less than 20-25)

Level of Needed Supports

level of support or assistance needed (rather than on IQ):

- ✓ Intermittent
- ✓ Limited
- ✓ Extensive
- ✓ Pervasive

Causes of Mental Retardation

- ➢ Genetic and Constitutional Factors chromosomal abnormalities
- ➢ Neurobiological influences

Emotional/Conduct Disorders

-"a condition exhibiting one or more of the following characteristics over a long period of time and to a marked degree that adversely affects a child's educational performance"

- – Inability to learn not explained by other factors
- – Inability to have interpersonal peer relationships
- – Inappropriate behavior or feelings under normal circumstances
- – Pervasive mood of depression or unhappiness
- – Tendency to develop physical symptoms or fears

Classification of Individuals with Emotional or Behavioral Disorders

Clinically derived classification systems

- The *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision* (DSM-V) provides criteria
- Diagnosis involves observation of behavior over time and across different settings

Statistically derived classification systems

- Involves categories of disordered behaviors
  - Externalizing behaviors
  - Internalizing behavior

DIMENSION OF PROBLEM BEHAVIOR

1. Conduct disorder
2. Socialized aggression
3. Attention problems/immaturity
4. Anxiety/withdrawal
5. Psychotic behavior
6. motor tension excess

Etiologies of Emotional or Behavioral Disorders

- Biological risk factors
- Psychosocial (environmental)

Characteristics of Students with Emotional or Behavioral Disorders

1. **Learning characteristics**
   - Range of intellectual abilities, chronic school failure, absenteeism, grade retention, school dropout
2. **Social characteristics**
   - Difficulty building and maintaining relationships, aggressive behavior, experience rejection, externalizing and internalizing behaviors
3. **Language/communication characteristics**
   - Deficits in the areas of pragmatics, receptive, and expressive language and limited or inappropriate language use.

Assessing Students with Emotional or Behavioral Disorders

1. Assessment strategies include:
2. Interviews with student, parents, and teachers
3. Examination of student records
4. Parent, teacher, and student rating scales
5. Observations in multiple natural settings
6. Medical evaluations
7. Standardized academic and IQ testing
8. Functional behavioral assessment (FBA)
9. Strength-based assessments
10. Other informal assessment strategies

Physical and Health Impairments

Physical disabilities are those that impair normal physical functioning. They may be:

- Orthopedic
- Neuromotor
- musculoskeletal impairments

Types of Physical Disability

1. **Cerebral Palsy**

2. **Spina Bifida**.
3. **Neuromotor Issues**
4. **Orthopedic Impairments**
5. **Neuromotor Issues**
6. **Orthopedic Impairments**
7. **Musculoskeletal disorder**

The following are guidelines when teaching a student with physical disabilities in the general classroom:

1. The school and classroom should be assessed by a physical or occupational therapist to determine its accessibility.
2. School arrangements such as ramp handrails, widening of sidewalks, and doorways, and adjustment of the heights of equipment might be necessary.
3. In the classroom, the teachers should make sure that there is ready access to all parts of the room, including centers and materials.
4. Sometimes classroom temperatures may have to be adjusted to accommodate a student's heath needs (Rosenberg, Westling, & McLeskey, 2011 p. 390).

Health impairments are diseases, illnesses, or conditions, such as asthma and epilepsy that require special care or attention and may impair learning and normal functioning. Students may be weak, tired, or in pain.

Students with other **health impairments** tend to have more absences due to their illness. Common health conditions that may classify a student as other health impaired are:

- Asthma
- HIV/AIDS
- Sickle-Cell Disease/anemia
- Epilepsy
- Cancer
- Type 1 (Juvenile) Diabetes
- Cystic Fibrosis/ Tourette Syndrome Rheumatic fever (Rosenberg, Westling, & McLeskey, 2011)

The following are guidelines when teaching a student with *other health impairments* in the general classroom:

1. Know the warning signals for students with conditions such as respiratory problems, heart conditions, or other chronic health problems and how to respond to students' needs.
2. Follow universal precautions to avoid contact with any communicable diseases.
3. Have emergency plans in case of an equipment failure, an emergency physical problem, or problems occurring due to natural disasters.
4. Be aware of routine treatments and who is responsible for carrying them out.
5. Know the medication the student takes, who is to administer it, and the possible side effects.
6. Know special nutritional needs such as dietary restrictions, special diets, or special eating procedures.
7. Know how much the student is expected to participate in self-managing his special physical or health needs.
8. It is essential that the student practice as much independence as possible (Rosenberg, Westling, & McLeskey, 2011 p. 390).

Severe and Multiple Disabilities

-Students with **Multiple impairments** have more than one disability in cognitive, physical and/or functional abilities. They typically require intensive intervention and supports for activities of daily living.
Multiple disabilities means a student has more than one serious disability which may affect mobility, behavior, emotion or sensory abilities. Some characteristic challenges of individuals with severe multiple impairments are:

- Limited communication or speech impairment
- Problems with physical mobility
- Cognitive impairments
- Tendency to forget skills through disuse;
- A need for support in major life activities (e.g., domestic, leisure, community use, vocational

Causes/etiology of Multiple Disabilities Having multiple disabilities means that a person has more than one disability. Causes can include:

- Chromosomal abnormalities
- Premature birth
- Difficulties after birth
- Poor development of the brain or spinal cord
- Infections
- Genetic disorders
- Injuries from accidents (1)

Multiple Disabilities Aren't All the SAME
Multiple disabilities gives two examples of possible combinations of disabilities:
1. Intellectual disability and blindness; and
2. Intellectual disability and orthopedic impairment

Developmental disabilities include

1. **Autism**
2. **behavior disorders**
3. **brain injury**
4. **cerebral palsy**
5. **Down syndrome**
6. **fetal alcohol syndrome**
7. **intellectual disability**
8. **spina bifida.**

Severe disabilities requires mental retardation but does not require an additional disability.

1. **Seizures**
2. **cerebral palsy**
3. **sensory loss**
4. **Hydrocephalus**
5. **scoliosis.**

Visual Impairments

Visual impairment is a term experts use to describe any kind of vision loss, whether it's someone who cannot see at all or someone who has partial vision loss. Usually caused by the following:

1. Injury
2. Congenital
3. Infection
4. Inherited

Common Eye Disorders and Diseases

1. **Myopia** -Nearsightedness, or myopia is the difficulty of seeing objects at a distance.
2. **Hyperopia,** or farsightedness, is when light entering the eye focuses behind the retina.
3. **Presbyopia** (loss of near vision with age)
4. **Cataract -**a cloudy area in the lens of your eye. Cataracts are very common as you get older.
5. **Diabetic Retinopathy**. a complication of diabetes, caused by high blood sugar levels damaging the back of the eye (retina)
6. **Amblyopia**. (also called lazy eye) is a type of poor vision that happens in just 1 eye.
7. **Strabismus**. a disorder in which both eyes do not line up in the same direction
8. **Colorblindness**
9. **Glaucoma** is an increase in pressure inside the eye

Hearing Impairments

Hearing impairment -whether permanent or fluctuating, that adversely affects a child's educational performance.

Deafness is defined as "a hearing impairment that is so severe that the child is impaired in processing linguistic information through hearing, with or without amplification."

Types of Hearing Loss

There are four basic types of hearing loss:

1. Conductive hearing loss

-occurs when sound waves are not transmitted effectively to the inner ear because of some interference.

2. **Sensorineural hearing loss**

With the sensorineural hearing loss, the damage lies in the inner ear, the auditory nerve, or both

3. **Central hearing loss**

-In central hearing loss, the problem lies in the central nervous system, at some point within the brain. Comprehending speech is a complex task. Some people can hear volume perfectly well, but have trouble understanding what is being said.

4. Mixed hearing loss

-occurs when an individual has both a conductive and sensorineural hearing loss in the same ear.

5. Unilateral hearing loss refers to hearing loss in one ear only which can range from mild to total hearing loss in that ear.

Strategies available to Deaf/hard of hearing students can be broken down into two general categories.

- Sign Language
- Speech/Oral Communication (refers to the use of speechreading and auditory cues)
- Hearing Aids

Disability- impairment or limitations

Handicap- degree of disadvantage

People First Language

People First Language refers to the individual first and the disability second.

-People with disabilities are - first and foremost –people

- It is about respect and dignity, and it puts the person-not the condition first.
- Say: **People with Disabilities**

Instead of: Mental Retardation

- Say: **Cognitive Disability**

Instead of: Mental Retardation

- Say: **He has autism**

Instead of: He is autistic

- Say: **Joan uses a wheelchair**

Instead of: Joan is in wheelchair

Say: **Bob has a mental health condition**

Instead of: Bob is mentally ill.

- Say: **Accessible Parking**

Instead of: Handicapped Parking

- Say: **He has autism**

Instead of: He is autistic

- Say: **Anthon has a disability**

Instead of: Anthon is crippled

- Say: **She has a learning disability (diagnosis)**

Instead of: She's learning disabled.

- Say: **Congenital disability**

Instead of: Birth defect

John B. Watson-Father of Behaviorism

- **Little Albert Experiment**
- Understanding fears, love, phobias and prejudice

Habituation - Decrease tendency to respond to stimuli that become more familiar

BEHAVIORAL LEARNING THEORY

- Operates on the principle of attached to a **S-R** also known as **Adhesive Principle**
- Prefers to concentrate on actual behavior
- Conclusions are based on Observations of external manifestations of Learning

Classical Conditioning by Ivan Pavlov

-Classical means "in an established manner."

-An individual learns when a previously **neutral stimulus is paired with an unconditioned stimulus** until the neutral stimulus evokes a conditioned response

Principles of Classical Conditioning

1. **Stimulus Generalization**
2. **Extinction**
3. **Excitation/Inhibition**
4. **Discrimination**
5. **Spontaneous Recovery**

Edward Lee Thorndike Connectionism

- Puts more emphasis on the response of the organism, not limiting himself to the association between the stimulus and the response

LAWS OF LEARNING

1. Laws of readiness
2. Law of Exercise
3. Law of Effect
4. Law of belongingness
5. Law of Association
6. Law of Multiple Response

7. Law of Frequency
8. Law of Contiguity
9. Law of requirement

Stages of Learning
- Acquisition
- Fluency
- Generalization
- Adaptation

Burrhus Frederic Skinner- Operant Conditioning

Stresses the consequence of behavior in order to learn.
-Proved that reinforcement is a powerful tool in shaping and controlling behavior inside and outside the classroom.
-**Operant Conditioning-** using pleasant and unpleasant consequences to control the occurrence of behavior

Principles of Operant Conditioning
1. **Shaping**
2. **Chaining**
3. **Extinction**

Schedules of Reinforcement
These four schedules of reinforcement are sometimes referred to as
- fixed-ratio
- variable-ratio
- fixed-interval
- variable-interval.

Kinds of reinforcer
- **Primary**
- **Secondary**
- **Positive**
- **Negative**

Purposive Theory – Edward Tolman
-Often referred to as Sign Learning Theory
-States that organisms learns by pursuing signs to a goal
-Stressed the relationship between stimuli rather than stimulus – response.
- Learning is **purposive** and **goal-directed**
- Cognitive mapping
- Latent learning
- Intervening variable
- Reinforcement is **not essential** for learning

Albert Bandura -Social Observational Learning Theory
- Vicarious Learning
- Bobo doll experiment
- People learn through **observation, simulation, modeling** which means watching (observing), another called a model and later **imitating** the model's behavior.
- Concentrates on the **power of example**

Four Phases of Modeling:
- **Attention** –
- **Retention** –.

- ❖ **Motor Reproduction Process**
- ❖ **Motivational Process** –

Cognitive Perspective

Contributory of **Gestalt Psychology** together with **Kurt Koffka** and **Max Wertheimer**

- ❖ **Gestalt Psychology** – shape of thoughts that looks at the human mind and behavior
- ❖ Gestalt means **form, figures, configuration**

(a complete shape)

- ❖ "The whole is more than the sum of it's part"

DIFFERENT GESTALT PRINCIPLES

1. Figure and Ground – our eye differentiates an object form its surrounding area. a form, silhouette, or shape is naturally perceived as **figure *(object),*** while the surrounding area is perceived as **ground *(background).***

-Balancing figure and ground can make the perceived image more clear.

2. Proximity – when objects close together, unity occurs.

3. Continuation

4. Similarity

5. **Closure**

6. The law of Pragnanz

**Wolfgang Kohler- Insight Learning and Problem Solving Theory **

- **Insight Learning** refers to the sudden realization of a solution of a problem.
- The capacity to discern the true nature of situation
- The **imaginative power to** see into and understand immediately
- **Gaining insight** is a gradual process of **exploring, analyzing, and structuring** perception until a solution is arrived at.

INFORMATION PROCESSING THEORY (Richard Shiffrin and Richard Atkinson)

The individual learns when:

1. the human mind takes **in information (encoding),**
2. **performs operation** in it, **stores the information (storage)**
3. and **retrieves** it when needed **(retrieval).**

KINDS OF RETRIEVAL

1. **Recall**
2. **Recognition**

Memory – the ability to store information so that it can be used at a later time.

INFORMATION PROCESSING THEORY (Atkinson)

1. Sensory Register
2. Short-Tern Memory
3. Long Term Memory

Kinds of long term Memory

1.Episodic

2. Semantic

3. Procedural

Declarative or Explicit Memory

Non-declarative or Implicit Memory

Cumulative Learning by Robert Gagne

- **Robert Gagne's Cumulative Learning** § -any task or skill can be broken down to simpler skills which can still be further broken down to more simple tasks or skills.

<u>Nine Events of Instruction</u>

1. Gaining Attention
2. Informing Learner of Objective/s
3. Recalling Prior Knowledge
4. Presenting Material
5. Providing Guided Learning
6. Eliciting Performance
7. Providing Feedback
8. Assessing Performance
9. Enhancing Retention and Transfer

MEANINGFUL RECEPTION THEORY – David Ausbel

Meaningful learning occurs when new experiences are **related to what a learner** already **knows.**

May occur through:

1**. Reception** -the learner actively associates the substances of new chains concepts and so forth with relevant components of previous learning

2. Rote learning -a memorization technique based on repetition. The idea is that one will be able to quickly recall the meaning of the material the more one repeats it.

3. **Discovery learning** refers to various instructional design models that engages students in learning through discovery

4 Processes of Meaningful Learning

1. **Derivative Subsumption**
2. **Correlative Subsumption**
3. **Superordinate Learning**
4. **Combinatorial Learning**

Instructional mechanism proposed by **Ausubel** is the use of **advance organizers** which help to link new learning material with existing related ideas.

1. Expository organizer
2. Comparative Organizer
3. Narrative

Bruner's Constructivist Theory

-proposes that learners construct their own knowledge and do this by organizing and categorizing information using a coding system.

Instrumental Conceptualism

Implies the idea is that students are more likely to remember concepts they discover on their own

Spiral Curriculum- learning is spread out over time rather than being concentrated in shorter periods. In a spiral curriculum, material is revisited repeatedly over months and across grades.

Learning Modes

1. Enactive
2. Iconic
3. Symbolic

Jean Piaget is known as one of the first theorists in constructivism

Constructivism is a learning theory which holds that knowledge is best gained through a process of reflection and active construction in the mind (Mascolo & Fischer, 2005).

Social Constructivism – Lev Vygotsky

Scaffolding - competent **assistance or support**

- **ZPD** child acquires **new skills and information** the level at which a child finds a task too difficult to complete alone, but which he can accomplish with the **assistance or support of an adult or older peer**

Zone of Proximal Development (ZPD)

- gap between actual and potential encounters.

- **Actual Development** – what children can do on their own
- **Potential Development** – what children can do with help

Sternberg's Theory of Successful Intelligence

Sternberg (1997) also stated that to be a successful intelligent person, one must combine and balance the three abilities:

- **Analytical** -Analytical intelligence involves analyzation (from the word itself), evaluation, judgment or comparison and contrast.

-his ability is "…valued in tests and in classroom

- **Creativity -** It is ability to pursue endless possibilities of thinking and imagination

-"one [that] goes beyond the range of unconventionality…"

- **Practicality** -practical intelligence refers to the ability to relate the learning or knowledge to the real world.

The **WICS model** is a possible common basis for identifying gifted individuals (Sternberg, 2003c).

WICS is an acronym standing for

- Wisdom
- Intelligence,
- Creativity
- Synthesized.

According to this model, wisdom, intelligence, and creativity are essential for the gifted leaders of the future

Creative Thinking

Dr. E. Paul Torrance (1915 – 2003) is called ***The Father of Creativity.***

-Torrance invented the **"Torrance Tests for Creative Thinking**

Components Of Creativity

1. **Fluency** – the ability to generate quantities of ideas
2. **Flexibility** – the ability to create different categories of ideas, and to perceive an idea from different points of view
3. **Originality** – the ability to generate new, different, and unique ideas that others are not likely to generate.
4. **Elaboration** – the ability to expand on an idea by embellishing it with details or the ability to create an intricate plan

Problem-solving skills

Problem-solving skills help you determine the source of a problem and find an effective solution.

In order to be effective at problem solving you are likely to need some other key skills, which include:

- **Creativity**. Problems are usually solved either intuitively or systematically.

-Intuition is used when no new knowledge is needed - you know enough to be able to make a quick decision and solve the problem, or you use common sense or experience to solve the problem
-complex problems or problems that you have not experienced before will likely require a more **systematic and logical approach to solve.**

- **Researching Skills.**
- **Team Working.**
- **Emotional intelligence**
- **Risk Management.**
- **Decision Making.**

MOTIVATION

- **Ability** refers to what an individual can do or is able to do and motivation (or lack of it) refers to what a person wants to do.

THE 5 PRIMARY MOTIVATION FACTORS

- **Fear**
- **Peer Pressure**
- **Pride**
- **Recognition**
- **Money**

Types of Motivation

1. **Extrinsic Motivation** – learners reason to work or study lies primarily outside. themselves.
2. **Intrinsic Motivation** – learners reason for learning resides primarily inside or upon them.

Need is a physiological deficiency that creates a condition of disequilibrium in the body.

David McClelland's Human Motivation Theory (Need Theory)

- allows you to identify people's motivating drivers.
- He identified three motivators that he believed we all have:
- a need for achievement
- a need for affiliation,
- a need for power.

Drive Theory Clark Hull

Drive is a condition of arousal or tension that motivates behavior.

- **Drives** most typically have been considered to involve physiological survival needs: hunger, thirst, sleep, pain, sex.
- A drive results from the **activation of a need.**

SELF DETERMINATION And MOTIVATION THEORY by Edward Deci

- We have the capacity to take risks or challenges that can enrich our lives and develop ourselves more.
- **Sense of Self-Determination Variables**
    1. Choices
    2. Threats and deadlines
    3. Controlling statements
    4. Extrinsic rewards
    5. Surveillance and evaluation

EXPECTANCIES and VALUES THEORY by John W. Atkinson

Variables that Affects Motivation:

1. **Expectancy –** People must believe that they can accomplish a task; that is, they should have an expectancy about what they want to achieve.

2.Value – People should likewise place an importance or value in what they are doing.

ATTRIBUTION THEORY by Bernard Weiner

Attributions pertain to people's various explanations for success and failure – their beliefs about what causes attributions.

- Dimensions underlying people's attributions. People can explain events in many different ways.

SPECIAL EDUCATION

The Council for Exceptional Children lists the following terms and definitions taken from the individuals with Disabilities Education Act**.**

Exceptional Students

The term "exceptional" has often been used to describe unusual, unique, or outstanding qualities of people or objects.

the term "exceptional" refers to students who learn and develop differently from most others or students who have:

- exceptional learning styles
- exceptional talents,
- exceptional behaviors.
- Exceptional students are those who fall outside of the normal range of development.

The term "**learners with special educational needs**" (LSEN) refers to learners who, for whatever reason, need additional help and support in their learning.

- **Persons with disabilities (PWDs),** according the UN Convention on the Rights of Persons With Disabilities, include those who have long-term physical, mental, intellectual or sensory impairments which in interaction with various barriers may hinder their full and effective participation in society on an equal basis

Barriers to healthcare

People with disability encounter a range of barriers when they attempt to access health care including:

- Attitudinal barriers
- Physical barriers
- Communication barriers
- Financial barriers

RA 5250 – An Act Establishing A Ten-Year Training Program For Teachers Of Special And Exceptional Children In The Philippines And Authorizing The Appropriation Of Funds Thereof.

RA. 7277 The Magna Carta for Disabled Persons was enacted for the primary reason that persons with disabilities have the same rights as other people.

The seven types of disabilities mentioned in RA No. 7277

1. psychosocial disability
2. disability due to chronic illness
3. learning disability
4. mental disability,
5. visual disability,
6. orthopedic disability
7. communication disability.

RA 9442 – An Act Amending Republic Act No. 7277, Otherwise Known As The "Magna Carta For Disabled Persons, And For Other Purposes".

-to provide persons with disability, the opportunity to participate fully into the mainstream of society by granting them at least twenty percent (20%) discount in all basic services.

RA 10754 IRR OF RA 10754 -AN ACT EXPANDING THE BENEFITS AND PRIVILEGES OF PERSONS WITH DISABILITY (PWD)

PRESIDENTIAL DECREE 603 - "THE CHILD AND YOUTH WELFARE CODE

BATAS PAMBANSA 232 "EDUCATION ACT OF 1982"

SENATE BILL NO. 1414 "INCLUSIVE EDUCATION FOR CHILDREN AND YOUTH WITH SPECIAL NEEDS ACT"

A PWD ID is a valid identification card issued to persons with disabilities. This card serves as a proof for availing of the benefits and privileges for PWDs.

What is the Validity of the PWD Card?
The PWD ID is valid for three years as stated in the National Council on Disability Affairs Administrative Order No. 001 series of 2021
PWD List of Disabilities in the Philippines
The Department of Health (DOH) considers the following types of disabilities as eligible for a PWD ID:

- Psychosocial disability –
- Disability resulting from a chronic illness - Includes orthopedic disability due to cancer, blindness due to diabetes, and other disabilities due to a chronic disease
- Learning disability
- Visual disability
- Orthopedic (Musculoskeletal) disability
- Mental/Intellectual disability
- Hearing disability
- Speech impairment
- Multiple disabilities

Intellectual Disabilities

Fragile X syndrome is a genetic condition that causes a range of developmental problems including learning disabilities and cognitive impairment.
Down Syndrome - Also known as Trisomy 21
-Genetic disorder caused by the presence of all or part of a third copy of chromosome 21.
A Developmental Delay refers to a child who has not gained the developmental skills expected of him or her, compared to others of the same age.
Delays may occur in the areas of :

- motor function
- speech and language,
- cognitive, play, and
- social skills

Prader-Willi syndrome is caused by some missing genetic material in a group of genes on chromosome number 15.
Fetal alcohol syndrome is a condition in a child that results from alcohol exposure during the mother's pregnancy.

Physical Disabilities

Cerebral palsy (CP) is a group of disorders that affect a person's ability to move and maintain balance and posture.
What are causes of cerebral palsy?
Causes

- Gene mutations that result in genetic disorders or differences in brain development.
- Maternal infections that affect the developing fetus.
- Fetal stroke, a disruption of blood supply to the developing brain.
- Bleeding into the brain in the womb or as a newborn.

Arthritis means inflammation or swelling of one or more joints.
A spinal cord injury — damage to any part of the spinal cord or nerves at the end of the spinal canal (cauda equina) — often causes permanent changes in strength, sensation and other body functions below the site of the injury.

Epilepsy is a central nervous system (neurological) disorder in which brain activity becomes abnormal, causing seizures or periods of unusual behavior, sensations and sometimes loss of awareness.

Muscular Dystrophy

Muscular dystrophies are a group of muscle diseases caused by mutations in a person's genes.

Emotional and Behavioral Disorders

Acute stress reaction occurs when a person experiences certain symptoms after a particularly stressful event.

Bipolar Disorder

-formerly called manic depression, is a mental health condition that causes **extreme** mood swings that include **emotional highs** (mania or hypomania) and **lows (depression).**

10 Signs of Bipolar Disorder

1. Decreased need for sleep. ...
2. Racing thoughts and accelerated speech. ...
3. Restlessness and agitation. ...
4. Overconfidence. ...
5. Impulsive and risky behavior. ...
6. Hopelessness....
7. Withdrawal from family and friends and lack of interest in activities. ...
8. Change in appetite and sleep.
9. **Conduct Disorder**

Conduct disorder refers to a group of behavioral and emotional problems characterized by a disregard for others.

There are four basic types of behavior that characterize conduct disorder:

- Physical aggression (such as cruelty toward animals, assault or rape).
- Violating others' rights (such as theft or vandalism) .
- Lying or manipulation.
- Delinquent behaviors (such as truancy or running away from home).

Eating Disorders

1. **Anorexia Nervosa**
2. **Bulimia Nervosa**
3. **Pica**
4. **Purging disorder**
5. **Night eating syndrome (NES)**

SPED Interventions

Shadow Teaching

-is when an educational paraprofessional, like a teaching aide or assistant, works directly with young students who have learning differences to improve their classroom experience.

Behavior Modification

- **Praise**
- **Rewards**
- **Behavior Chart**
- **Redirection**
- **Engage**
- **Visuals**

- **Extinction**
- **Direct Instruction**

  What are the four steps of direct instruction?

  Direct Instruction guides us through intermediary stages to gently transition from teacher to student.

1. **Modeling – The teacher does it all.**
2. **Structured Practice – The teacher does it, but with input from students.**
3. **Guided Practice – Students do it, with input from the teacher.**
4. **Independent Practice – Students do it.**